Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming 1st part–January 22nd, 2026

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and ensure to provide a brief explanation. Remember, without a reason, the answer will be considered invalid. Understanding the questions is crucial for your success in this exam.

A. What kind of synchronization does PRAM use? (1)

B. If the PRAM model is a theoretical model, why does it matter in real problem analysis? (1)

C. During the lectures, we discuss how to compute the sum of a collection of integer values in a PRAM model. Could you discuss the conclusions we reached regarding the problem's complexity? (1)

D. Please outline the key aspects of Gustafson's law and how it relates to scalability. (1)

Exercise n. 2

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss why, for many algorithms we have discussed during the lectures, the complexity of the best sorting algorithm is bounded by $O(n \lg n)$. (2)

B. Please explain why counting sort matters with the radix sort algorithm. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

C. Describe how the split for Treaps data structure works and discuss its complexity. (1)

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Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Could you outline the main differences between Las Vegas and Monte Carlo algorithms, especially in terms of efficiency, and give examples of each? (1)

B. Please describe an example of an online problem and how its complexity could be analyzed. (1)

C. Please describe a problem that is typically solved by a divide-and-conquer strategy, yet also admits a dynamic-programming formulation. In your explanation, highlight how memoization is employed. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Could you explain which aspects of primality testing we have seen during the lectures that use randomization? Additionally, would you classify it as a Monte Carlo or Las Vegas randomized algorithm? (2)

B. Explain the various forms of amortized analysis. What characteristics of a problem make amortized analysis the appropriate analytical technique? (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**